

G-CAF

GovDigital Composable Architecture Framework

Volume 6

Implementation Guide

v2.0

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Chapter 1

Implementation Journey Overview

G-CAF implementation is supported by 18 documents organized across four priority tiers, guiding the complete journey from pilot to government-wide adoption.

Priority 1: Critical for Launch

Document	Module ID	Purpose
Entity Onboarding Playbook	G-CAF-IMPL-001	Step-by-step entity adoption guide
Change Management Strategy	G-CAF-IMPL-002	Organizational change approach
Communication Plan	G-CAF-IMPL-003	Stakeholder communication strategy
Pilot Selection Criteria	G-CAF-IMPL-004	Objective criteria for pilot selection

Priority 2: Important for Sustainability

Document	Module ID	Purpose
Benefits Realization Framework	G-CAF-IMPL-005	Track and demonstrate ROI
Success Metrics Dashboard	G-CAF-IMPL-006	Consolidated KPIs and measurement
Resource & Capacity Model	G-CAF-IMPL-007	Resource requirements guidance
Risk Register	G-CAF-IMPL-008	Implementation risks and mitigations

Priority 3: Operational Excellence

Document	Module ID	Purpose
Tooling Strategy	G-CAF-IMPL-009	IDP, Catalog, Portal requirements
Integration Playbook	G-CAF-IMPL-010	Integration with PMO, Procurement, IV&V
Vendor Transition Guide	G-CAF-IMPL-011	Vendor adaptation to G-CAF
Compliance & Audit Framework	G-CAF-IMPL-012	Audit evidence requirements

Priority 4: Enablement & Adoption

Document	Module ID	Purpose
Quick Start Guides by Role	G-CAF-IMPL-013	Persona-specific guides
Maturity Model	G-CAF-IMPL-014	Comprehensive maturity assessment

Document	Module ID	Purpose
Community of Practice Charter	G-CAF-IMPL-015	CoP establishment guide
Community Activation Playbook	G-CAF-IMPL-016	Cultural principles, contributor ladder
Contributor Ladder	G-CAF-IMPL-017	Individual progression L0–L5

Chapter 2

Entity Onboarding Playbook

The onboarding playbook guides each entity through a 5-stage journey from initial awareness to self-sufficient G-CAF operations.

Onboarding Stages

Stage	Duration	Key Milestone	Outputs
1. Awareness	2 weeks	Executive sponsor confirmed	Sponsor assigned, team identified, kickoff completed
2. Assessment	4 weeks	Readiness score complete	Maturity score, gap analysis, priority PBCs identified
3. Planning	4 weeks	Adoption roadmap approved	Roadmap, resource plan, training plan
4. Execution	12–24 weeks	First PBC through Go-Live	PBCs in Catalog, gateways passed, contracts approved
5. Optimization	Ongoing	Self-sufficient operations	Reuse rate >50%, CoP participation

Total time to first PBC: 22–34 weeks (5–8 months).

Support Levels

Wave	Entities	Support Level	Approach
Wave 1 (Months 1–6)	3–5 entities	High-touch (dedicated EA)	Daily check-ins, on-site workshops
Wave 2 (Months 4–12)	10–15 entities	Medium-touch (shared EA)	Weekly check-ins, virtual workshops
Wave 3 (Months 10–18)	20–30 entities	Low-touch (self-service)	Office hours, documentation, CoP

Chapter 3

Change Management Strategy

G-CAF represents a fundamental transformation across architecture, governance, operations, and culture for 55+ entities and 10,000+ IT professionals over 3–5 years.

Transformation Dimensions

Dimension	From (Current)	To (Target)
Architecture	Siloed systems, point-to-point	Composable PBCs, standardized contracts
Governance	Ad-hoc oversight, inconsistent	Gateway-based, standard contracts, DDA/ARB
Operations	Entity-owned, manual, reactive	Shared platforms, IDP automation, proactive SLAs
Culture	Build-first, vendor dependency, project-focused	Run-first, government ownership, product-focused

Scale of Change

Dimension	Impact
Scope	55+ government entities, ~10,000+ IT professionals
Depth	Fundamental architecture shift affecting all roles
Duration	3–5 year sustained transformation
Complexity	High — technical + organizational concurrent changes

Chapter 4

Pilot Selection Criteria

Pilot Objectives

Objective	Success Measure
Prove the concept	First PBCs through all gateways successfully
Refine the process	Gateway process validated and improved
Build capability	Trained architects and teams
Demonstrate value	Quantified benefits (time, cost, quality)
Create advocates	Entity sponsors become G-CAF champions
Generate momentum	Success stories drive broader adoption

Entity Selection Criteria

Entities are scored across: Executive Sponsorship (25%), Architecture Readiness (20%), IT Maturity (15%), Willingness to Change (15%), PBC Candidates (15%), and Visibility (10%).

Pilot Waves

Wave	Timeline	Entities	PBCs	Focus
Wave 1	Months 1–6	3–5	3–5 (1 per entity)	Prove concept, low-risk PBCs
Wave 2	Months 4–12	10–15	15–25 (1–2 per entity)	Scale validation, moderate complexity

Chapter 5

Resource & Capacity Model

Capacity Planning by Phase

Phase	Timeline	Central FTE	Entity FTE (each)	Focus
Foundation	Months 1–6	15–20	2–3	Setup, pilots
Scale	Months 7–18	25–30	3–5	Broad rollout
Optimize	Months 19–36	20–25	2–4	Efficiency, self-service
Steady State	Ongoing	15–20	2–3	Maintenance, evolution

Central Resources

Team	Key Roles	Focus
EA Office	Chief EA, Domain Architects, Framework Team	Architecture governance, standards
Platform Team	Platform Engineers, DevOps, IDP Specialists	IDP, pipelines, infrastructure
Governance	DDA Secretariat, VRO Analysts, Training Team	Gateway process, value tracking, training

Entity Resources

Role	Key Responsibilities
Entity Architect	G-CAF alignment, solution architecture, gateway preparation
Solution Architects	PBC design, contract implementation, integration patterns
PBC Owners	PBC lifecycle, business alignment, value delivery
Developers	PBC implementation, testing, golden path compliance
PMO	Project governance, resource coordination, reporting

Chapter 6

Risk Register

Risk Categories

Category	Key Risks
Strategic	Executive sponsorship loss, strategy misalignment, competing priorities, funding gaps
Organizational	Resource shortages, skills gaps, resistance to change, change fatigue
Technical	Platform complexity, integration challenges, data quality issues, tool maturity
Operational	Governance process bottlenecks, support capacity, training delivery
External	Vendor dependency, regulatory changes, market shifts, procurement delays
Benefit	ROI not realized, reuse not achieved, benefits delayed

Risk Score Interpretation

Score Range	Risk Level	Response
20–25	Critical	Immediate executive action, contingency plans
15–19	High	Active management, regular monitoring
10–14	Medium	Monitor, standard mitigation
5–9	Low	Accept, periodic review
1–4	Very Low	Accept

Chapter 7

G-CAF Maturity Model

5 Maturity Levels

Level	Name	Description
Level 1	Initial	Ad-hoc, inconsistent, individual-dependent
Level 2	Developing	Basic processes, some consistency, reactive
Level 3	Defined	Standardized processes, consistent practices, documented
Level 4	Managed	Quantitatively managed, predictable outcomes, proactive
Level 5	Optimizing	Continuous improvement, innovation, industry-leading

6 Assessment Domains

Domain	Description	Weight
Governance	Gateway process, decision-making, compliance	20%
Architecture	PBC design, contracts, standards adherence	25%
Delivery	Speed, quality, predictability	15%
Operations	Reliability, support, monitoring	15%
Reuse	Catalog usage, consumption, contribution	15%
Capability	Skills, training, culture	10%

Chapter 8

Benefits Realization Framework

Benefits Categories

Category	Examples
Strategic	Digital Strategy alignment, technical debt control, governance maturity, innovation enablement
Operational	Delivery acceleration, quality improvement, reuse efficiency, process standardization
Financial	Cost reduction (30–50%), cost avoidance, investment protection, TCO optimization
Risk	Security posture improvement, compliance confidence, vendor independence, change resilience
Capability	Skills development, architecture capability, modern practices, collaboration culture

Target Benefits

Benefit	Key Metric	Target	Timeline
Delivery Acceleration	Time to deliver	50–70% faster	Per PBC
Cost Reduction	Development cost	30–50% reduction	Annual
Reuse Rate	PBC reuse	Over 50%	Quarterly
Quality Improvement	Defect rate	40% reduction	Quarterly
Governance Efficiency	Gateway cycle time	Within SLA	Monthly

Chapter 9

Success Metrics Dashboard

5-Level Metrics Hierarchy

Level	Audience	Frequency	Focus
Level 1: Strategic	Digital Board	Quarterly	Government-wide adoption, total benefits (AED), strategic alignment
Level 2: Program	ARB/TMO	Monthly	Entity adoption status, PBC portfolio, governance effectiveness
Level 3: Operational	DDA/EA Office	Weekly	Gateway pipeline, training, support metrics, quality
Level 4: Entity	Entity CIO	Monthly	Entity-specific progress, PBC metrics, entity benefits
Level 5: Community	CoP Council	Quarterly	Contributor ladder distribution, entity diversity, cross-entity reuse

Chapter 10

Tooling Strategy

Tool Categories

Category	Tools	Purpose
Developer Experience	IDP, CI/CD Pipelines, Schema Registry	Self-service, golden paths, templates, automated builds
Governance & Catalog	PBC Catalog, DDA Portal, Knowledge Base	Discovery, registration, gateway workflow, documentation
Operations	Observability, Security Scanning, Infrastructure as Code	Monitoring, logging, tracing, SAST/DAST, Kubernetes

Tool Priorities

Priority	Tools	Timeline	Investment
P1: Critical	IDP Core, PBC Catalog, DDA Portal, CI/CD	Year 1	High
P2: Important	Schema Registry, Observability, Security Scanning	Year 1-2	Medium
P3: Enablers	Knowledge Base, Advanced Analytics, Self-service	Year 2-3	Medium
P4: Optimization	AI-assisted tooling, Advanced automation	Year 3+	Low

Chapter 11

Compliance & Audit Framework

7 Compliance Domains

Domain	Key Areas
Governance Compliance	Gateway process adherence, DDA decisions, approvals, escalations
Architecture Compliance	PBC design principles, interface contracts, catalog registration
Operations Compliance	SLA compliance, operational readiness, monitoring, incident response
Security Compliance	Authentication, encryption, access controls, audit logging, privacy
Data Compliance	Data ownership, classification, privacy, retention policies
Vendor Compliance	Contract compliance, SLA achievement, gateway participation
Benefits Compliance	Value Frame tracking, benefits validation, reuse measurement

Compliance Levels

Level	Description	Applies To
Mandatory	Must comply; blocking at gateways	Gateway requirements, security
Expected	Should comply; tracked and reported	Documentation, processes
Recommended	Best practice; encouraged but not enforced	Optimization, maturity improvements

Chapter 12

Community of Practice & Quick Start Guides

Community of Practice (CoP)

A vibrant community of G-CAF practitioners who learn from each other, share best practices, and continuously improve composable architecture across Abu Dhabi Government.

The CoP is organized as a council overseeing specialized guilds (Architecture, Integration, Platform), each with working groups tackling specific challenges.

Contributor Ladder

Level	Role	Activities
L0	Observer	Attends events, reads documentation
L1	Learner	Completes training, asks questions in forums
L2	Practitioner	Delivers PBCs, shares lessons learned
L3	Contributor	Submits RFCs, mentors others, leads working groups
L4	Expert	Recognized authority, shapes framework direction
L5	Fellow	Governs CoP, represents externally, drives innovation

Vendor Transition

Technology vendors and system integrators must understand and adapt to G-CAF requirements. The Vendor Transition Guide covers core requirements (interface contracts, schema registration, gateway participation), vendor types (Platform Vendors, SaaS Providers, System Integrators, Custom Developers, Managed Service Providers), and the benefits of compliance (clear requirements, predictable process, cross-entity reach, longer relationships).

Quick Start Guides by Role

Role	Key Knowledge	First Actions
Entity CIO	G-CAF in 60 seconds, executive role	Assign sponsor, allocate resources, track PBC metrics
Entity Architect	3T+2F model, classification, gateways	Assess readiness, identify first PBC, prepare for G1
Solution Architect	PBC design, contracts, interface framework	Design PBC, implement contracts, pass G2
PBC Owner	PBC lifecycle, value frame, catalog	Register PBC, manage through gateways, track value
Developer	Golden paths, API standards, CI/CD	Use templates, implement contracts, pass automated checks

Role	Key Knowledge	First Actions
PMO	Gateway process, resource planning, Coordination	Coordinate gateways, track milestones, manage risks
Procurement	DDA Technical Committee, vendor requirements	Include G-CAF in RFIs, evaluate vendor compliance
Vendor	G-CAF requirements, gateway participation	Implement interface contracts, register APIs, support SLAs

Department of Government Enablement (DGE) | Abu Dhabi
Enterprise Architecture Office | govdigital.ea@dge.gov.ae